Problem set 4

PPHA 31102 Statistics for Data Analysis II: Regressions

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1. Trade Policy

Tariffs frequently impact the flow of goods internationally. In this section, you will draw on data from one of these tariffs, investigating how these disruptions impact exports. For the next five questions, you will use the data posted on Canvas as *trade_policy.csv*. This data contains several variables: post is a binary variable indicating the tariff has been introduced (post = 1, pre = 0); *treat* is a binary variable that equals 1 if a country is "impacted" by the tariff barrier (i.e., a regulation increasing the cost of exports) and 0 if a country is "not impacted"; *air_conditioners_pc* is the number of air conditioners exported per capita by a given country in a year.

Question 1.1

Use a difference-in-differences design to estimate the impact of the tariff on exports of air conditioners. Regress air_conditioners_pc on post, treat, and the interaction of post and treat. Which coefficient reveals the change in exports after the tariff was introduced? Explain and interpret the result.

```
tp$interaction <- tp$treat * tp$post
model <- lm(air_conditioners_pc ~ treat + post + interaction, data = tp)
stargazer(model, type="text",  out="models.txt")
##
## ================================================
##                          Dependent variable:
##                      ---------------------------
##                          air_conditioners_pc
## ------------------------------------------------
## treat                          -0.228**
##                                 (0.090)
##
## post                           0.286***
```

```
##                                (0.098)
##
## interaction                   -0.634***
##                                (0.128)
##
## Constant                      0.964***
##                                (0.069)
##
## -----------------------------------------------
## Observations                    3,280
## R2                              0.029
## Adjusted R2                     0.028
## Residual Std. Error      1.809 (df = 3276)
## F Statistic          32.092*** (df = 3; 3276)
## ===============================================
## Note:               *p<0.1; **p<0.05; ***p<0.01
```

Question 1.2 (2 points)

A friend is suggesting that you should consider taking the log of your outcome, air_conditioners_pc. What are some potential concerns regarding this suggestion? Describe two.

Answer: [Your answer here]

Question 1.3 (2 points)

The difference-in-differences design is valid when treatment and control have common trends. What is a common trend? Write down the definition. Next, give an example (about this case) when this assumption is likely violated.

Answer: [Your answer here]

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2. The Oregon Health Insurance Experiment, Revisited (16 points)

```
# Load the OHIE data
```

Question 2.1 (2 points)

Consider the variable ever_medicaid. Did everyone who won the lottery take up Medicaid? Calculate the mean of ever_medicaid among those who won the lottery. Are there some *people who lost the lottery who take up Medicaid? Calculate the mean of ever_medicaid for those that lost the lottery. Is the causal effect of winning the lottery likely to be the same as the causal effect of actually receiving Medicaid?*

```
# Mean of ever_medicaid among lottery winners

# Mean of ever_medicaid among lottery losers
```

Answer: [Your interpretation here]

Question 2.2 (2 points)

Estimate the "First-stage" regression:

$$ever\_medicaid = \alpha 0 + \alpha 1 * treated + \alpha 2 * numhh_list + e$$

Interpret the regression coefficient on $\alpha 1$.

```
# First-stage regression
```

Answer: [Your interpretation here]

Question 2.3 (3 points)

Is treated likely to be a valid instrument for ever_medicaid in terms of the three key assumptions: instrument relevance, independence assumption, and the exclusion restriction?

Answer: [Your answer here]

Question 2.4 (2 points)

Calculate and report the ratio of the reduced form to the first stage for the outcome count_visit_dr. Note: use the same observations (exclude rows with missing count_visit_dr).

```
# Subset to non-missing count_visit_dr

# Reduced form regression

# First-stage regression (same sample)

# Calculate ratio
```

Answer: [Your interpretation here]

Question 2.5 (3 points)

Manually estimate the two stages of 2SLS. How does your result compare to the previous estimate? Is the IV estimate statistically significant?

```
# (a) First-stage and predicted values

# (b) Second-stage regression
```

Answer: [Your interpretation here]

Question 2.6 (2 points)

Re-estimate the 2SLS model including the full set of controls (numhh_list, female, age, age^2, race_white, hs_degree, college_degree, health_baseline). Compare the coefficient on ever_medicaid and the standard error to the results in part (5).

```
# Create age squared variable

# 2SLS with full controls
```

Answer: [Your comparison here]

Question 2.7 (2 points)

For each endline outcome, run the IV regressions including controls using ivreg(). Fill in the table and discuss findings.

```
# IV regression for visit_dr

# IV regression for visit_er

# IV regression for out_of_pocket_spend

# IV regression for health_score

# IV regression for happy
```

Endline outcome	(1) IV estimate	(2) S.E.
visit_dr		
visit_er		
out_of_pocket_spend		
health_score		
happy		

Answer: [Your discussion here]